Design review #3 – Space graded INA

Project developed by:	**SLICE Semiconductor**
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1 General information

This project will develop an instrumentation amplifier (INA) with programmable gain (through I2C / SPI interfaces) which achieves high accuracy(< 1% gain error), low noise (< 50nV/rtHz at 1kHz), and high common mode rejection (> 80dB). It will produce an application specific integrated circuit (ASIC) using free and open tools which will be taped out on IHP's 130nm open source PDK, with performance guaranteed across process (slow / fast silicon), voltage (± 5%) and temperature. Because of its high performance, the INA will be applicable to high accuracy sensing applications (pressure, strain, temperature) typical to those found in a wide variety of industry / medical applications. In addition, the INA will be made robust to high radiation environments, making it applicable to low earth orbit / high energy particle applications, where single even latch up can be problematic. The final chip will be characterised as a tangible proof of the maturity of today's free and open source toolchains and their readiness to produce high performance semiconductors.

1.1 Project goal

To show the capability of open source tools to develop industry quality ASICs, verified across industry standard conditions and taped out on silicon from a European fab (IHP).

To validate this goal, targeted performance will match that of an INA from a leading industry vendor (AD8223A [1]).

1.2 Specifications

Below specifications to be met across full process (SS/FF/SF/FS), voltage (±5%) and temperature (-40 → 125C) (i.e. PVT) conditions.

Parameter	Unit	Value	Comment
vdda_hv	V	3.3	Analog supply
vdda_lv	V	1.5	Digital supply
vssa	V	0	Ground
idda_hv	mA	1	Analog supply current
Power	mW	3.465	Analog power consumption
Gain error	%	0.5	INA Accuracy within ±50mV of the supply
|CMRR|	dB	80	Common mode rejection ratio
Noise density	nV/rtHz	50	Measured at 1kHz
Integrated noise	uVrms	200	Set by temp sense requirements
Bandwidth	kHz	50	
LG(0)min	dB	70	For all -ve feedback loops
PM min	deg	55	10deg > post layout spec to give margin
|GM| min	dB	13	3dB > post layout spec to give margin
Noise density	nV/rtHz	30	For all -ve feedback loops
|CMRR(0)| min	dB	120	Applicable to pamps in all -ve feedback loops
idda_hv	uA	300	For all -ve feedback loops

Table 1: INA specifications

1.3 Purpose of Design review #3 (DR3)

Purpose of DR3 is to review final layouts before delivery to top level (i.e. chip level) integration. Therefore, upon approval of DR3, layouts are “frozen” and top level integration commences. Specifically, this DR3 refers to the layout freeze of the INA top level (layout freezes for blocks outside of the INA top level are detailed in separate DR3’s).

1.4 Repository details

All material relating to this project can be found in the below publicly available repository:

https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main

Screenshot of the project landing page is shown in Figure 1.

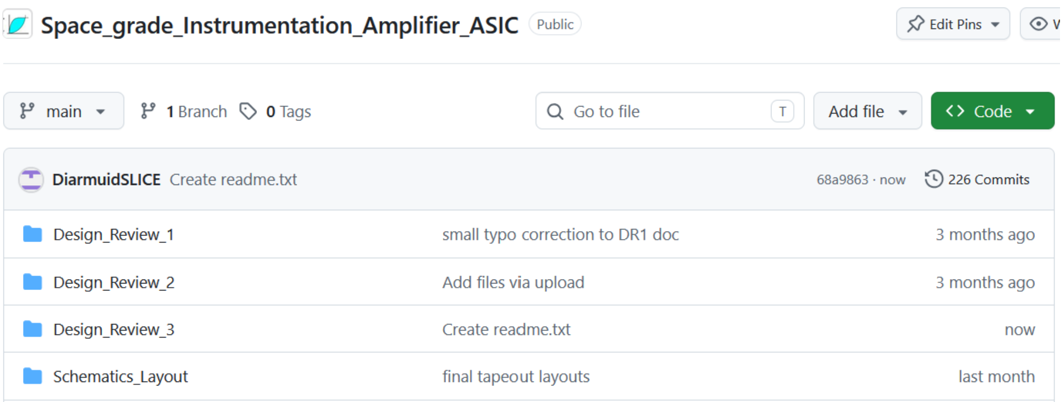

Figure 1 Project landing page

As this document pertains to DR3, all material relating to it is found in the sub directory “Design_Review_3”. As block schematics relate to all DR’s, these are contained in the separate folder “Schematics_Layout”. As shown in Figure 2 this folder contains a sub-directory to each block in the ASIC design.

Note: Figure 2 shows only a small portion of the total number of blocks contained in this folder.

Inside each folder here, lives the following views:

- Schematic (to be frozen at DR2)
- Symbol (to be frozen at DR2)
- Layout (to be frozen at DR3)

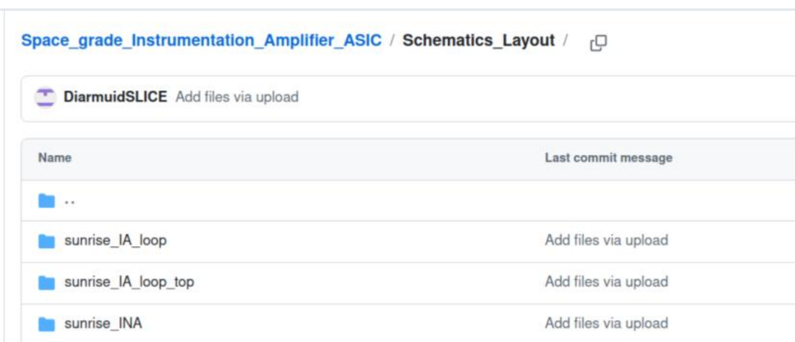

Figure 2 Project design blocks

1.5 Open source tools

The open source tools used for DR3 are summarized in Table 2, with links to install each tool provided as references.

Tool	Description
ngspice	Spice simulator [2]
xschem	Schematic / waveform viewer [3]
openVAF	verilogA compiler [4]
Octave	Post processing [5]
klayout	Layout viewer / editor [6]
magic	Pcell generator / PEX generation [7]
netgen	LVS [8]

Table 2: Open source tools used for DR3

1.6 Open source PDK

The open source process development kit (PDK) used for the design is from IHP, process option – SG13CMOS5L [9]. This is a derivative of their more general option – SG13G2 [10] as it excludes the following:

- Top metal layer 2
- Linear caps (MiM or MoM)
- Deep Nwell (DNW)
- HBT devices

The first 3 items impacted the design but not the 4th since the design is not RF. SG13CMOS5L option was chosen primarily because its tapeout date of 13/07/26 suited the project timeline. In addition, it is lower cost (max = €1.5k/mm2 as opposed to €2.8k/mm2 for SG13G2) and runs though the fab quicker (5 months instead of 6) to achieve a fab out date in 16/12/26 [11].

2 Top-Level design description

2.1 Architecture

Top level architecture was thoroughly detailed in section 2.1 of DR1 [12], where the INA was shown to comprise of 3 -ve feedback loops, repeated in Figure 3 for reference.

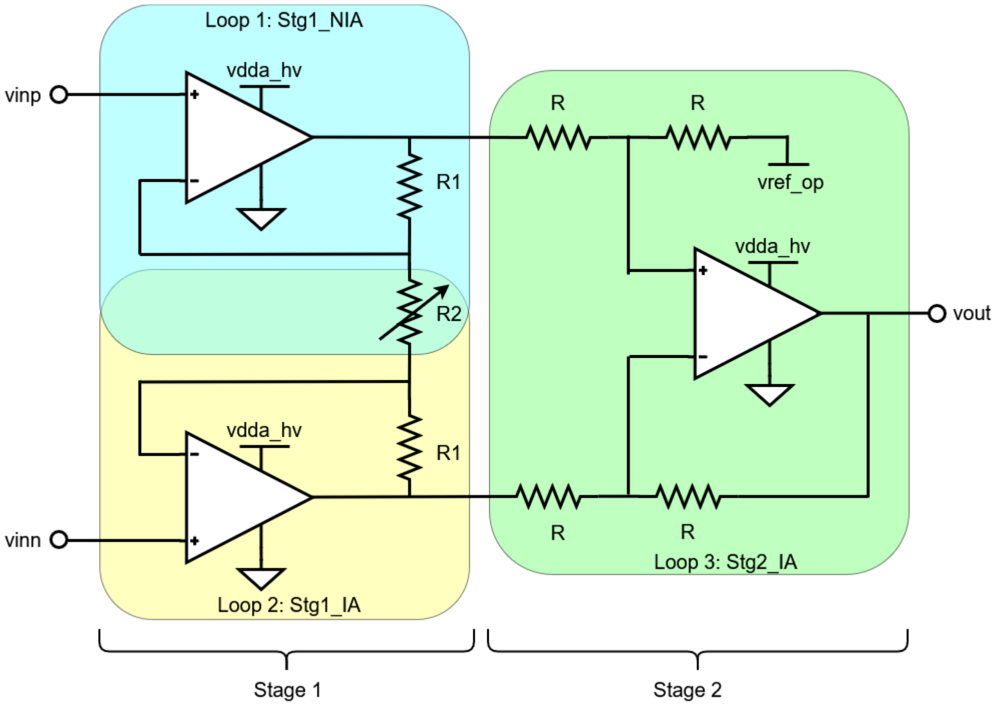

Figure 3 INA top level architecture

For layout efficiency, the same 2-stage opamp was used for each INA stage whose details were provided in section 2.1 of DR2 [13].

As detailed in DR1, the INA can be configured into 4 different gain settings – 1 / 5 / 11 / 21, where functionality / performance is guaranteed by design across PVT for gain settings 5 / 11 / 21, and functionality only guaranteed for the unity gain setting. This is because INA's are never used in unity gain since all output referred error sources translate directly back to the input (i.e. they don't get attenuated when referred back to the input) and is in keeping with industry standards where datasheets rarely provide performance at unity gain.

3 Testbench description

3.1 Testbench details

To verify all layouts, a 1:1 comparison of simulation results is made between the layout for the block under test and its corresponding schematic, in the same testbench. This is achieved by simply replacing the schematic view of the block under test, with a parasitic extracted (PEX) netlist describing the corresponding layout.

This means the testbenches detailed in section 3.1 of DR2 (to verify schematics) are re-used in DR3, to verify their corresponding layouts. Reader is thus referred to DR2 for testbench details and how to run them.

4 Design hierarchy

For ease of layout and physical verification (PV) checks, design has been constructed into the multi-layer hierarchy summarized in Table 2.

1	2	3	4	5	6	7	8	9
sunrise_INA_top	sunrise_mux	sunrise_tgate						
	sunrise_INA_wbias	sunrise_bias	sunrise_bias_ctrl	sunrise_tgate_small				
			sunrise_opamp_bias					
		sunrise_INA	sunrise_NIA_loop_top	sunrise_NIA_loop	sunrise_res_100k			
					sunrise_opamp_top	sunrise_opamp_cmfb_prb		
							sunrise_opamp_bias	
							sunrise_opamp_stg1	
							sunrise_opamp_stg2	
								sunrise_opamp_pcomp
								sunrise_opamp_ncomp
			sunrise_IA_loop_top	sunrise_IA_loop	sunrise_res_100k			
					sunrise_opamp_top	sunrise_opamp_cmfb_prb		
							sunrise_opamp_bias	
							sunrise_opamp_stg1	
							sunrise_opamp_stg2	
								sunrise_opamp_pcomp
								sunrise_opamp_ncomp
			sunrise_STG2_loop_top	sunrise_STG2_loop	sunrise_res_10k			
					sunrise_opamp_top_ndppd	sunrise_opamp_cmfb_prb_ndppd		
							sunrise_opamp_bias	
							sunrise_opamp_stg1_ndppd	
							sunrise_opamp_stg2	
								sunrise_opamp_pcomp
								sunrise_opamp_ncomp
			sunrise_rdac	sunrise_res_100k				
				sunrise_tgate				
	sunrise_cfilt							

Table 2 Design hierarchy

INA top level symbol is shown in Figure 4.

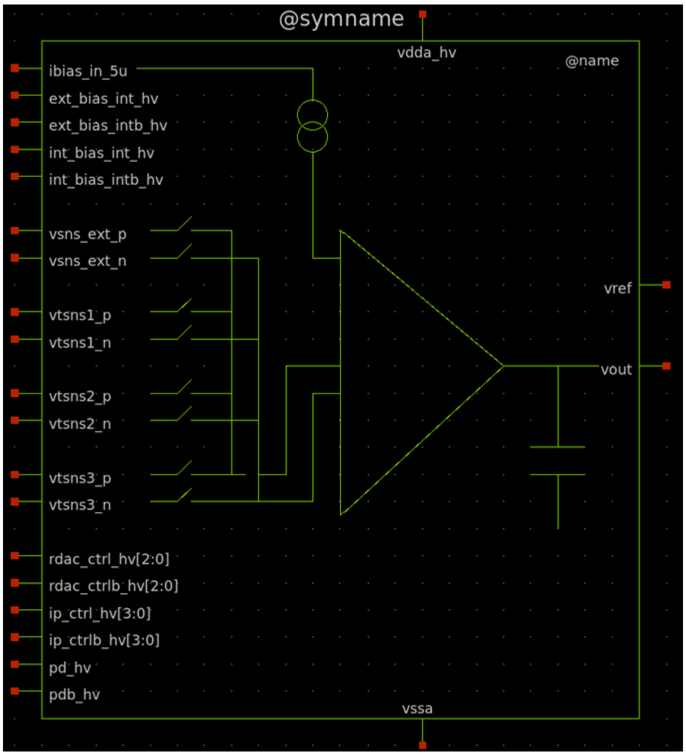

Figure 4 INA top level symbol

5 Testbench simulation results (sub-blocks)

5.1 Stage 1 NIA loop

5.1.1 Layout view

Stage 1 NIA loop schematic is shown in Figure 5 for reference. Its corresponding layout is shown in Figure 6 where it can be seen to occupy a footprint of ~ 430 x 470um.

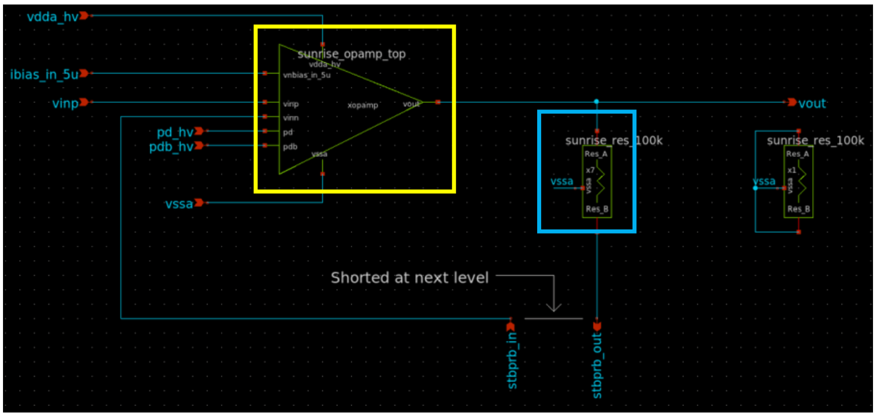

Figure 5 Stg1 NIA loop schematic

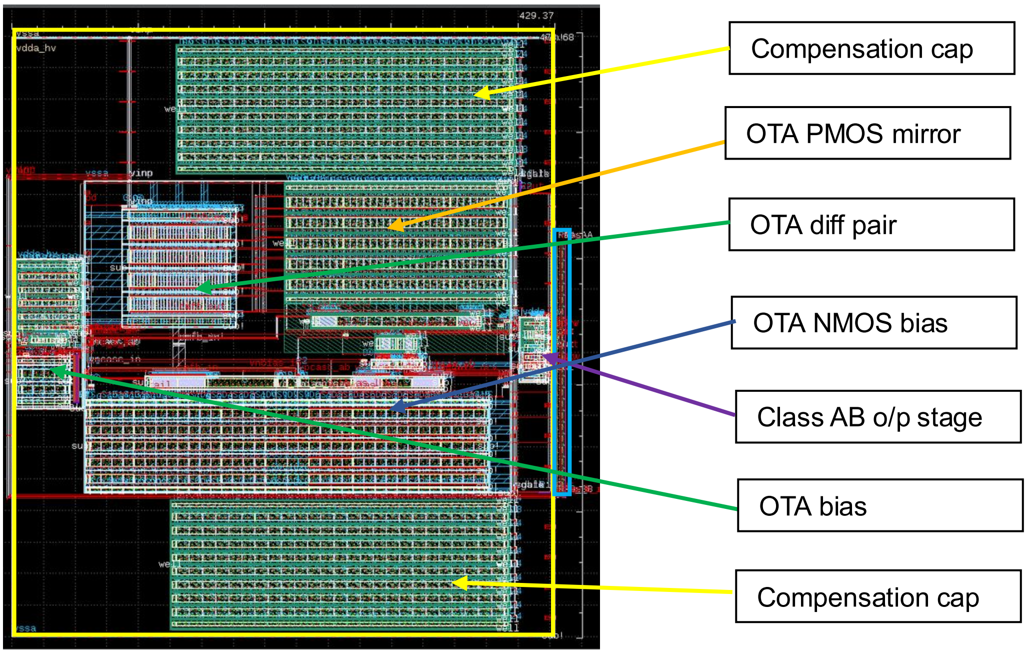

Figure 6 Stage 1 NIA loop layout

It is worth nothing the close abutment of the feedback resistor to the opamp output. This is to eliminate offset / headroom issues which arise from placing this resistor too far from the opamp output. As shown in Figure 7, it is placed ~ 10um from the output to avoid headroom issues, with connections to / from the feedback resistor occurring directly on the top and bottom of the resistors to avoid any offset.

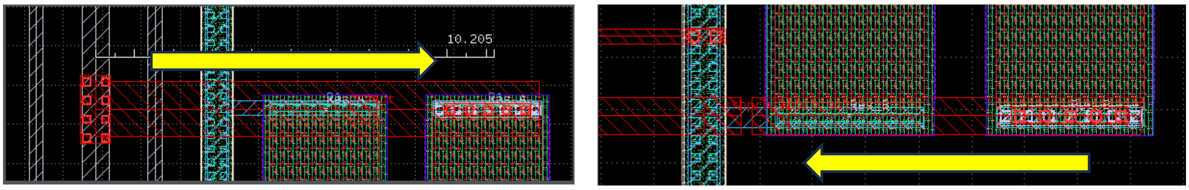

Figure 7 Connections to and from the feedback resistor

All major blocks are delivered with a full power grid on M4/M5. Figure 8 is Figure 6 but with the power grid included.

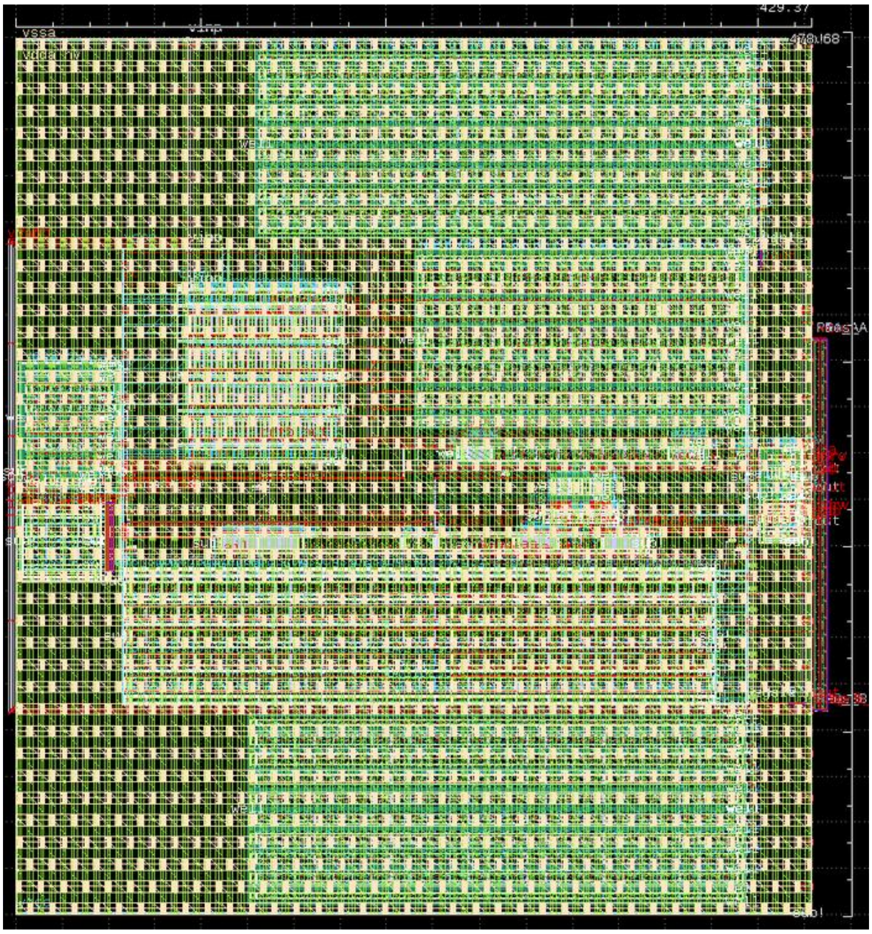

Figure 8 Stage 1 NIA layout with power grid

It is worth noting that the RDAC (R2 in Figure 3) was also included in post layout analysis, whose schematics and corresponding layouts are shown in Figure 9 and Figure 10 respectively.

It is worth noting from Figure 10 how all resistors occur in the same bank. This is to minimize mismatch that otherwise arise by placing them in different locations / at different orientations from each other.

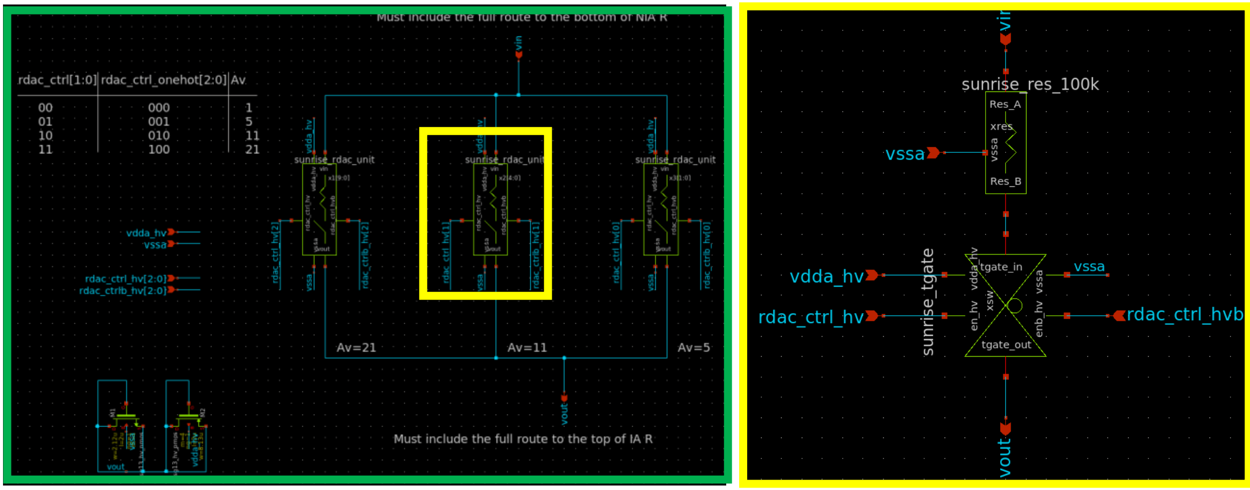

Figure 9 RDAC schematic

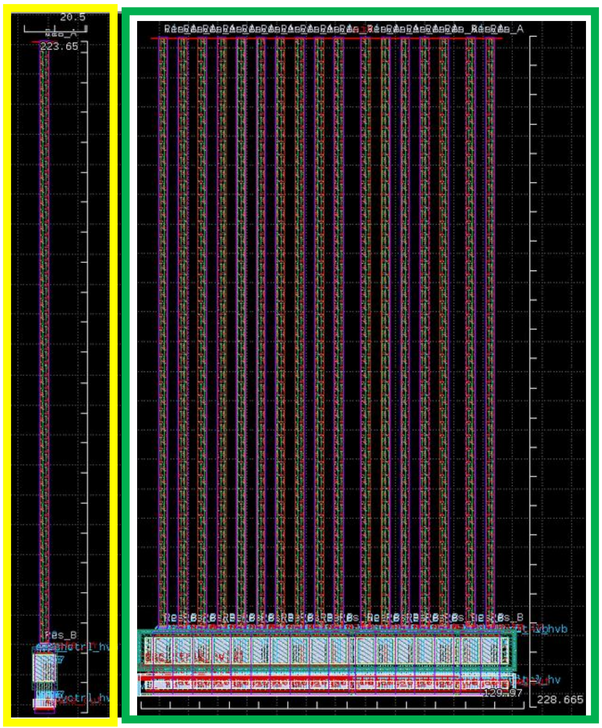

Figure 10 RDAC layout unit and top level schematic

5.1.2 Stability

Stability analysis is summarized in Table 3 and plotted in Figure 11.

Parameter	SCH	PEX	Delta
LG_0_min (dB)	108	108	0
ULGF (MHz)	64.1	60.5	3.6
PM_min (deg)	68.8	66.4	2.4
|GM|_min (dB)	27.1	25.8	1.3

Table 3 Stability summary

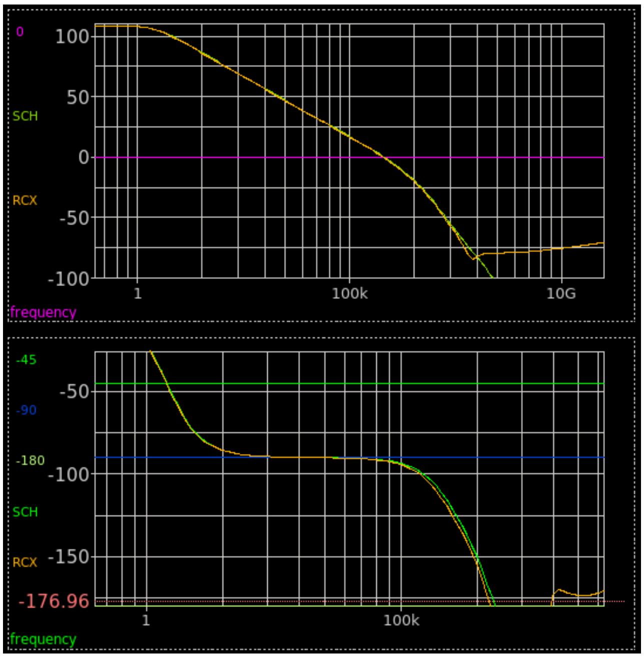

Figure 11 Pre (SCH) and post (RCX) layout stb plots

As can be seen, stability performance pre and post layout match very closely. This took numerous layout iterations to achieve, specifically to adjust the compensation caps to offset the impact of parasitic caps from the layout.

As these results correspond to open loop stability in the frequency domain, the closed loop stability of the loop in the time domain needs to be run to fully validate the stability of the entire loop. This is performed by analysing the loops response to a 10mV step applied at the input. As shown in Figure 12, step responses pre and post layout match closely, in agreement with the open loop stability analysis.

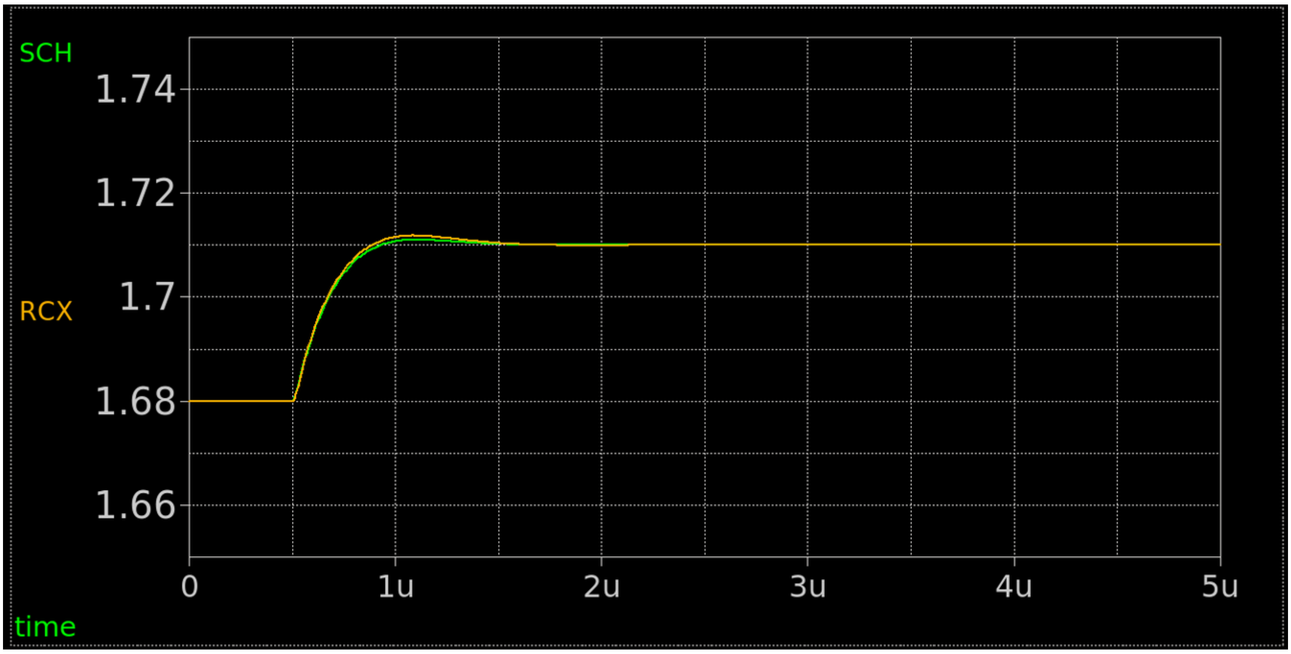

Figure 12 Pre (SCH) and post (RCX) layout step responses

5.1.3 Noise density

Input referred noise density at 1kHz is summarized in Table 4 with its corresponding output referred noise density plotted in Figure 13.

Parameter	SCH	PEX	Delta
vn_1kHz (V/rtHz)	22.6	22.6	0

Table 4 Noise density summary

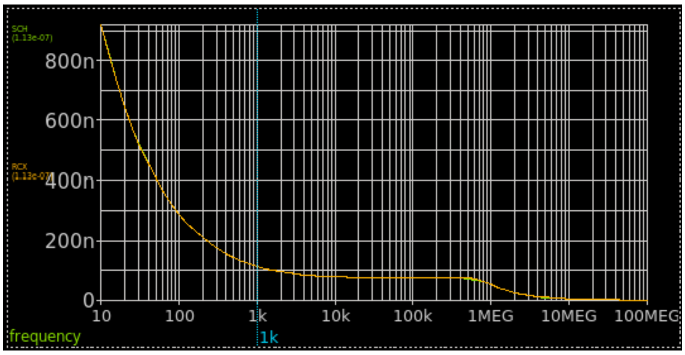

Figure 13 Pre (SCH) and post (RCX) layout noise density plots

As can be seen, there is no difference pre and post layout which is to be expected at 1kHz.

5.1.4 idda_hv

Current consumption is summarised in Table 5.

Parameter	SCH	PEX	Delta
idda_hv (uA)	254	254	0

Table 5 idda_hv summary

As can be seen, there is no difference pre and post layout.

5.1.5 Post layout summary of stage 1 NIA loop

Pre and post layout analysis for stage 1 NIA loop is summarized in Table 6, verifying the layout to match schematic closely.

Parameter	SCH	PEX	Delta
LG_0_min (dB)	108	108	0
ULGF (MHz)	64.1	60.5	3.6
PM_min (deg)	68.8	66.4	2.4
|GM|_min (dB)	27.1	25.8	1.3
vn_1kHz (V/rtHz)	22.6	22.6	0
idda_hv (uA)	254	254	0

Table 6 Pre and post layout summary for the stage 1 NIA loop

5.2 Stage 1 IA loop

Looking closely at Figure 3 one can see that both stage 1 loops are physically the same, which becomes obvious when viewing the top level INA schematic in Figure 14. While both loops are differentiated for explanation / reporting reasons, physically speaking they are the same and hence realized with the same layout. As such, the analysis in section 5.1.1 covers this section, where it will not be repeated.

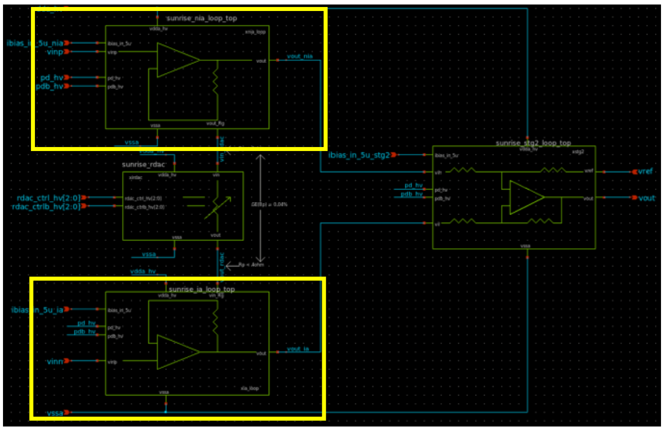

Figure 14 INA top level schematic

5.2.1 Stability

Stability analysis is summarized in Table 7 and plotted in Figure 15.

Parameter	SCH	PEX	Delta
LG_0_min (dB)	108	108	0
ULGF (MHz)	64.8	60.1	4.7
PM_min (deg)	70.4	65.9	4.5
|GM|_min (dB)	23.6	22.7	0.9

Table 7 Stability summary

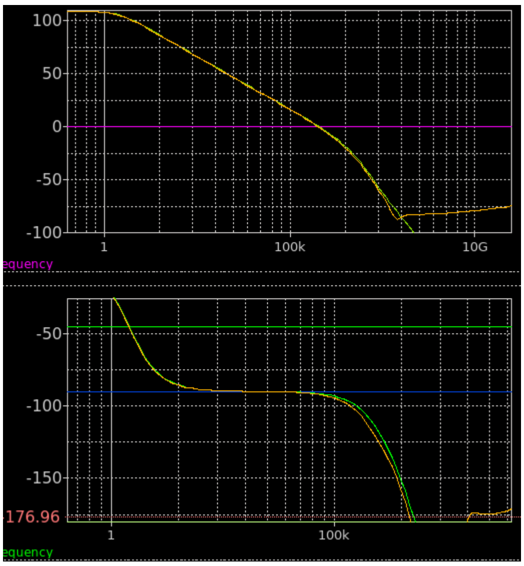

Figure 15 Pre (SCH) and post (RCX) layout stb plots

As shown in Figure 16, step responses pre and post layout match closely, in agreement with the open loop stability analysis.

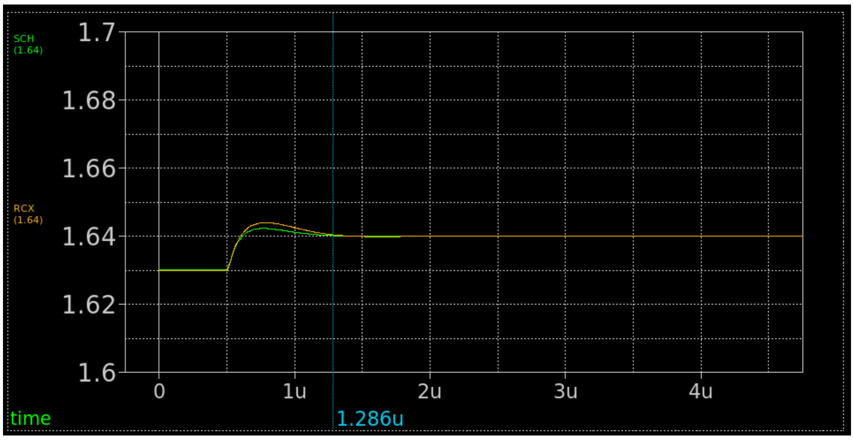

Figure 16 Pre (SCH) and post (RCX) layout step responses

5.2.2 Noise density

Input referred noise density at 1kHz is summarized in Table 8 with its corresponding output referred noise density plotted in Figure 17.

Parameter	SCH	PEX	Delta
vn_1kHz (V/rtHz)	22.6	22.6	0

Table 8 Noise density summary

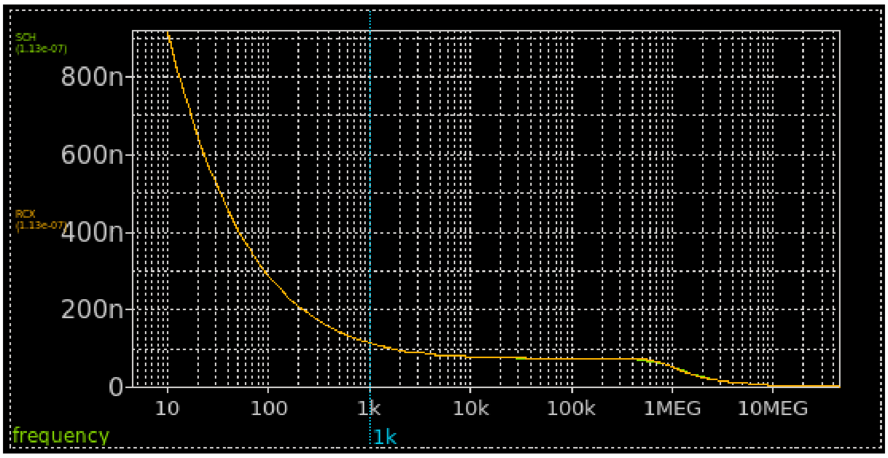

Figure 17 Pre (SCH) and post (RCX) layout noise density plots

As can be seen, there is no difference pre and post layout which is to be expected at 1kHz.

5.2.3 Post layout summary of stage 1 IA loop

Pre and post layout analysis for stage 1 NIA loop is summarized in Table 9, verifying the layout to match schematic closely.

Parameter	SCH	PEX	Delta
LG_0_min (dB)	108	108	0
ULGF (MHz)	64.8	60.1	4.7
PM_min (deg)	70.4	65.9	4.5
|GM|_min (dB)	23.6	22.7	0.9
vn_1kHz (V/rtHz)	22.6	22.6	0

Table 9 Pre and post layout summary for the stage 1 IA loop

5.3 Stage 2 IA loop

5.3.1 Layout view

Stage 2 IA loop schematic is shown in Figure 18 for reference with its corresponding layout shown in Figure 19. Excluding the resistors, it occupies the same 430 x 470um since the opamp is almost identical (only change being 2 pd switches on the diff pair inputs).

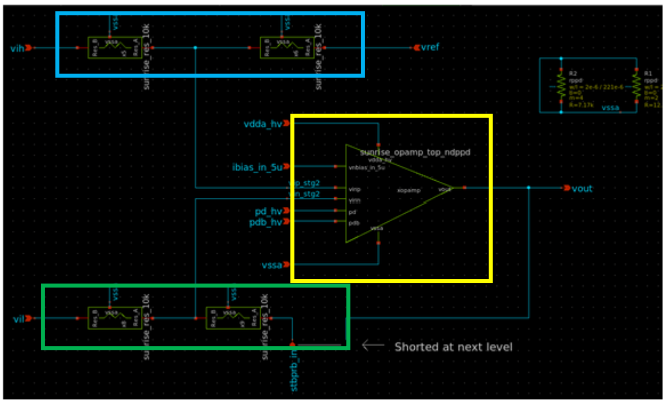

Figure 18 Stg2 IA loop schematic

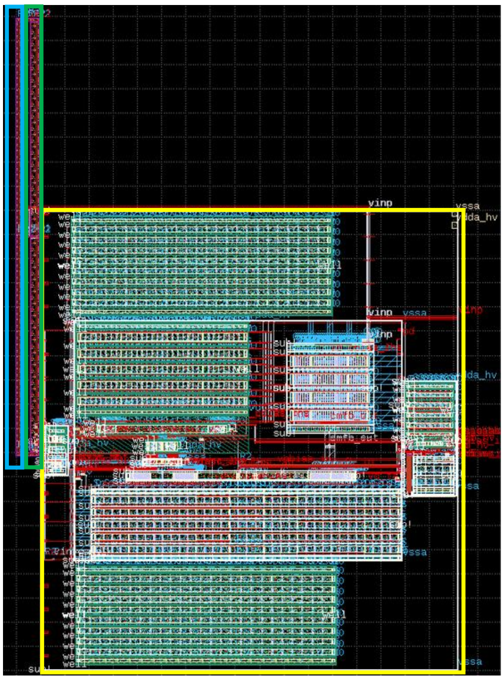

Figure 19 Stage 2 IA loop layout

It is worth nothing the close abutment of the resistors to the opamp output. This is to eliminate offset issues which arise from placing this resistor too far from the opamp output. As detailed on slide 14 in [14], these resistors were originally placed along the top of the opamp to yield a more square shaped layout. However, the resulting offset degraded CMRR by 60dB due to its common mode dependance, necessitating the resistors to be placed next to the opamp output as shown in Figure 20.

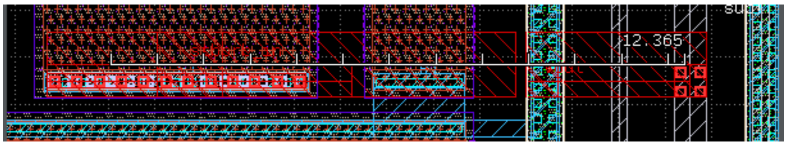

Figure 20 Connection from opamp output to the resistors

As with the stage 1 loop, this block was delivered with a full power grid on M4/M5. Figure 21 is Figure 19 but with the power grid included.

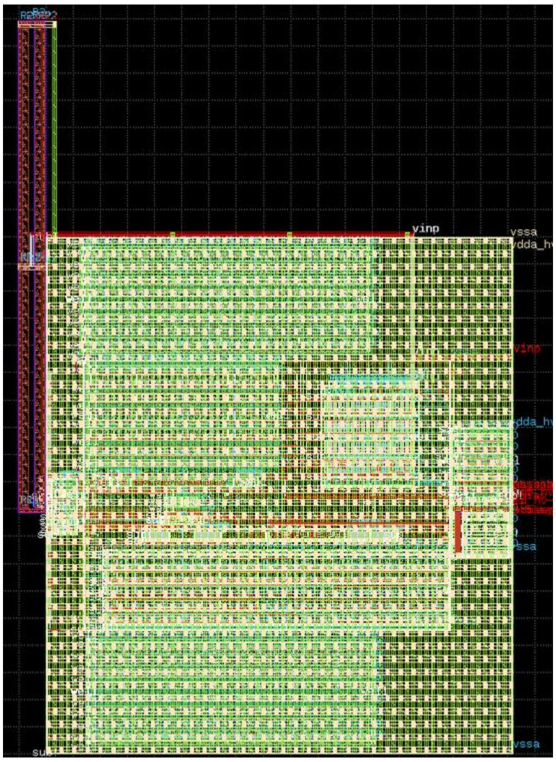

Figure 21 Stage 2 IA loop layout with power grid

5.3.2 Stability

Stability analysis is summarized in Table 10 and plotted in Figure 22.

Parameter	SCH	PEX	Delta
LG_0_min (dB)	101	101	0
ULGF (MHz)	107.2	105.7	1.5
PM_min (deg)	91.4	92.1	-0.7
|GM|_min (dB)	22.2	18.8	3.4

Table 10 Stability summary

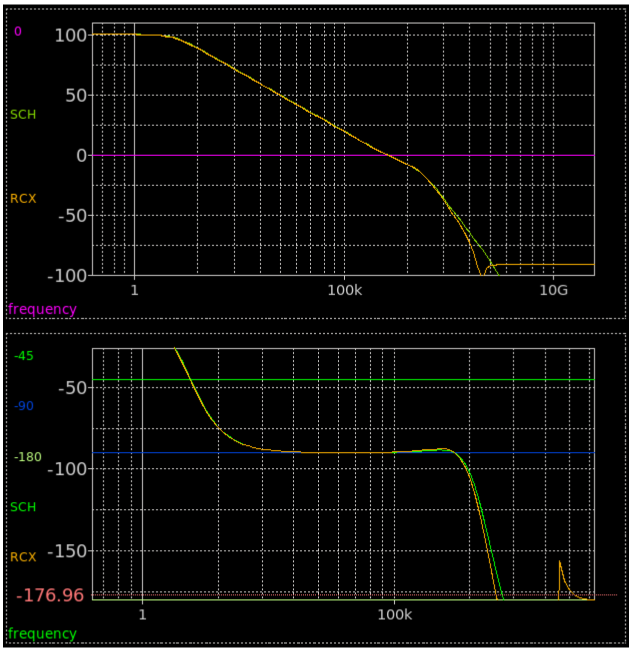

Figure 22 Pre (SCH) and post (RCX) layout stb plots

As shown in Figure 23, step responses pre and post layout match closely, in agreement with the open loop stability analysis.

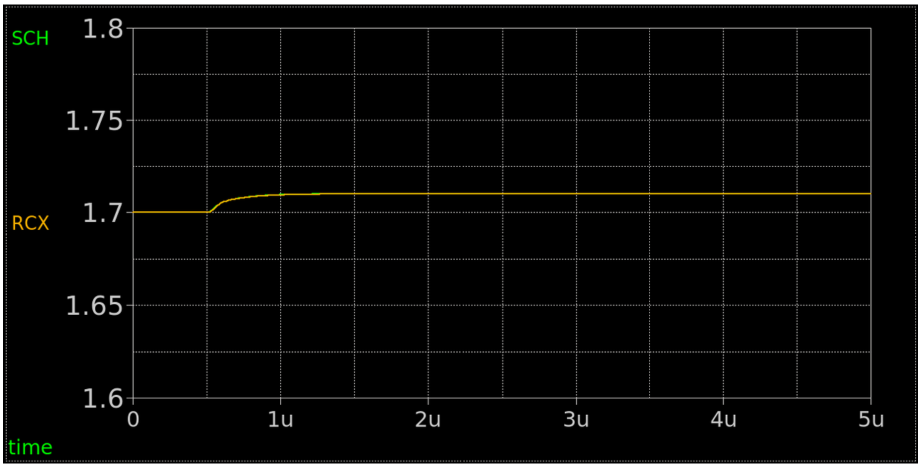

Figure 23 Pre (SCH) and post (RCX) layout step responses

5.3.3 Noise density

Input referred noise density at 1kHz is summarized in Table 11 with its corresponding output referred noise density plotted in Figure 24.

Parameter	SCH	PEX	Delta
vn_1kHz (V/rtHz)	12.9	12.9	0

Table 11 Noise density summary

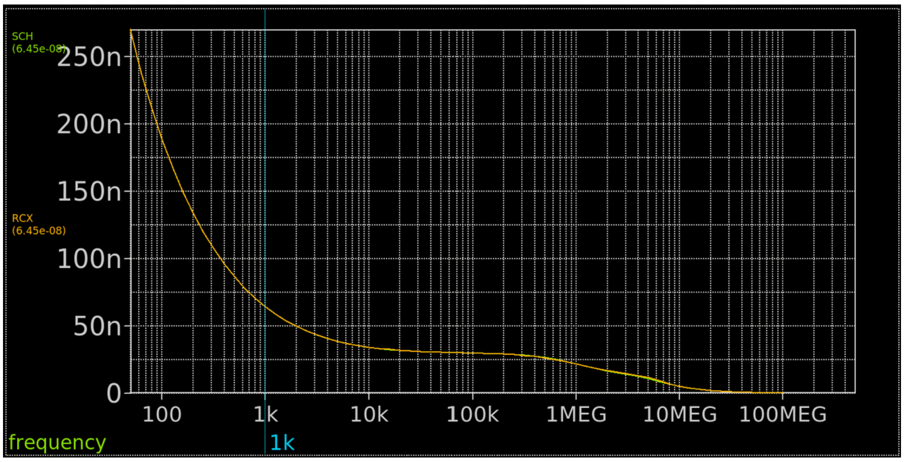

Figure 24 Pre (SCH) and post (RCX) layout noise density plots

As can be seen, there is no difference pre and post layout which is to be expected at 1kHz.

5.3.4 Post layout summary of stage 2 IA loop

Pre and post layout analysis for stage 1 NIA loop is summarized in Table 12, verifying the layout to match schematic closely.

Parameter	SCH	PEX	Delta
LG_0_min (dB)	101	101	0
ULGF (MHz)	107.2	105.7	1.5
PM_min (deg)	91.4	92.1	-0.7
|GM|_min (dB)	22.2	18.8	3.4
vn_1kHz (V/rtHz)	12.9	12.9	0

Table 12 Pre and post layout summary for the stage 2 IA loop

5.4 Tgates

In DR2 it was reported that 2 flavors of Tgate would be used. For layout efficiency this was later changed to one with no loss of performance. Furthermore, the schematic of that Tgate was optimized for layout (e.g. PMOS and NMOS with equal channel lengths and number of fingers). Resulting schematic is shown in Figure 25.

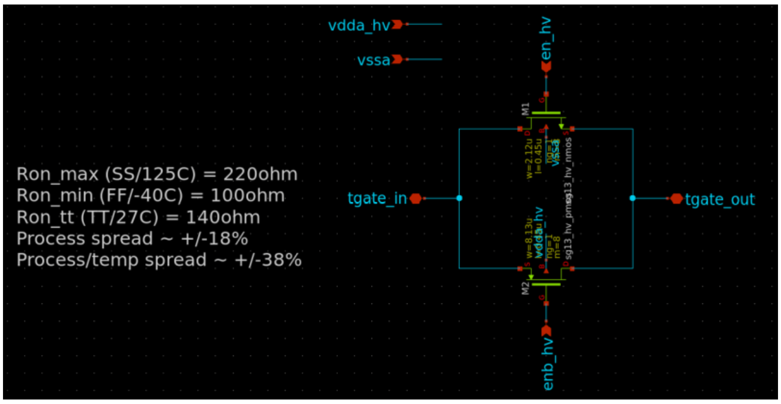

Figure 25 Schematic of the Tgate used throughout the design

Resulting layout is shown in Figure 26 where the advantage of employing the same number of fingers / channel lengths for each device can be clearly seen.

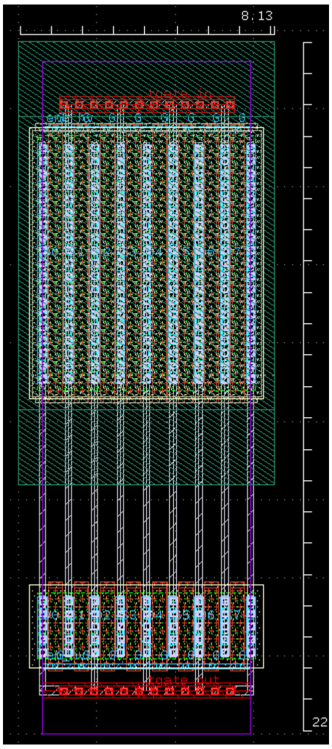

Figure 26 Layout of the Tgate used throughout the design

Ron (Rsw) as a function of the input voltage pre and post is shown in Figure 27, verifying the layout to match schematic closely.

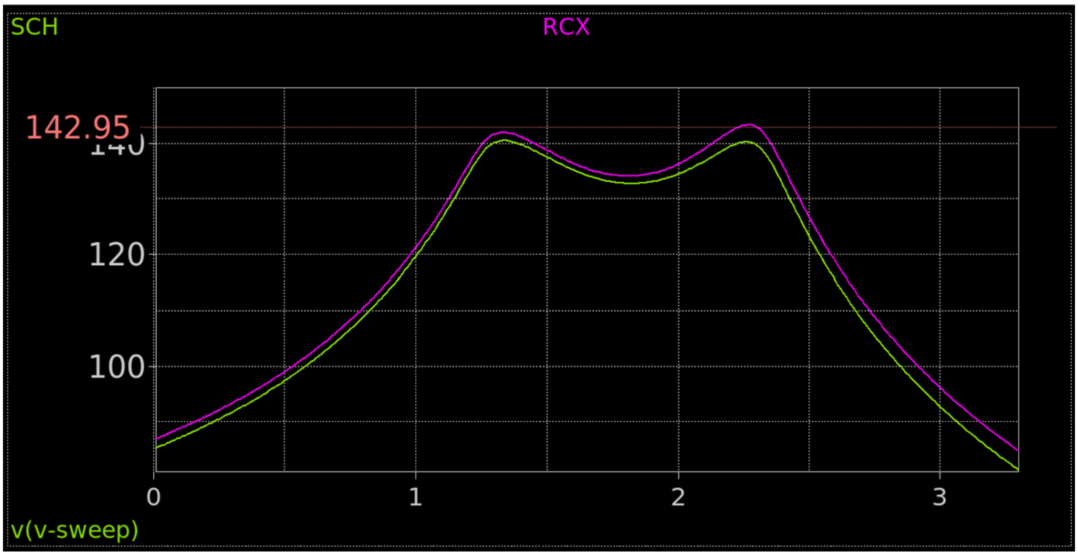

Figure 27 Tgate Ron pre (SCH) and post (RCX) layout

5.5 INA biasing

The INA takes a single 5uA current and distributes it to each of the 3 opamps. As shown in Figure 28, the INA biasing block responsible for this comprises of 2 sub-blocks – an input mux to pass through an internally or externally generated 5uA, and a bias block to mirror this out to each of the 3 opamps.

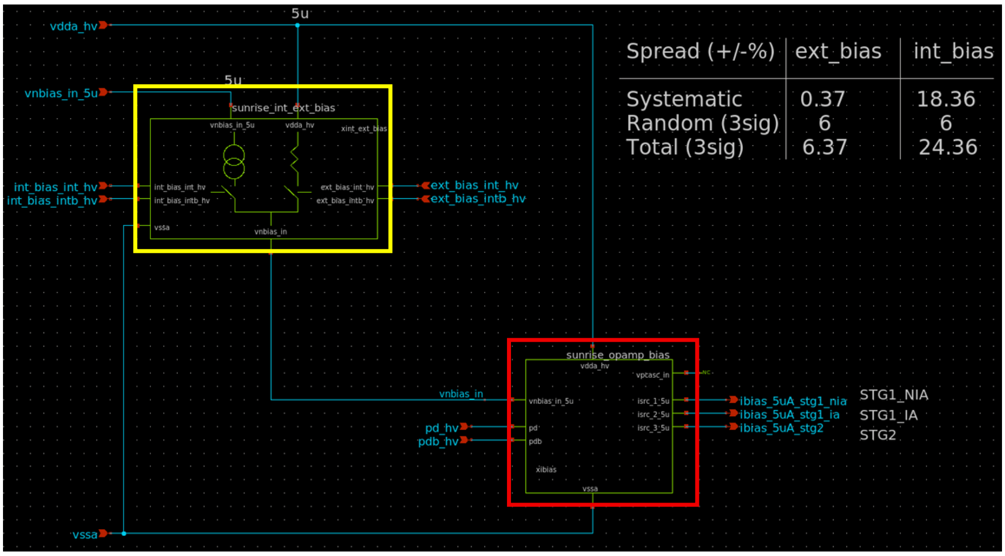

Figure 28 INA biasing schematic

Although the block used for this purpose in DR2 had the same functionality, its hierarchy differed slightly. For layout efficiency it was decided to re-use the biasing block within each of the opamps to mirror the input 5uA. As a result, from Figure 28, one can see output “vpcasc_in” tied to a no conn since, although used in the opamps, is not required in this block. Resulting layout is shown in Figure 29.

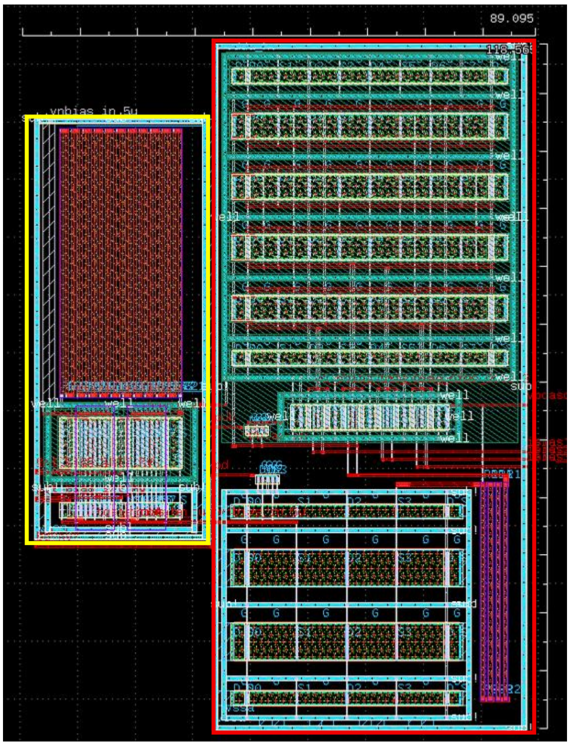

Figure 29 INA biasing layout

Comparing Figure 29 with Figure 6 one can clearly see the re-use of the bias block. Comparing with Figure 25 shows input mux to simply comprise of an abutment of 2 Tgates.

Output current from the block pre and post layout is shown in Figure 30, verifying the layout to match schematic very closely.

Figure 30 Output currents pre (SCH) and post (RCX) layout from INA biasing

6 Testbench simulation results (top level)

6.1 Layout view

Top level schematic is shown in Figure 31 for reference. Its corresponding layout is shown in Figure 32 where it can be seen to occupy a footprint of ~ 1 x 1mm.

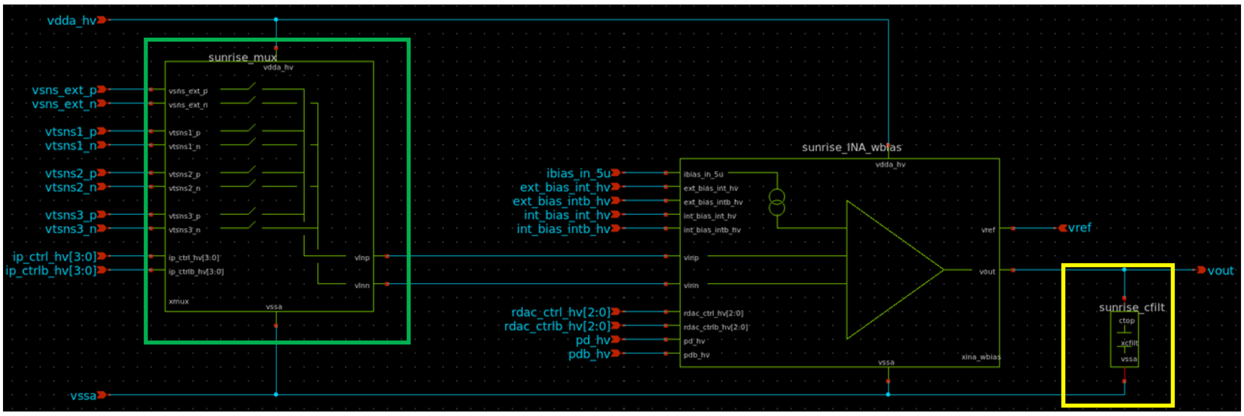

Figure 31 sunrise_INA_top schematic

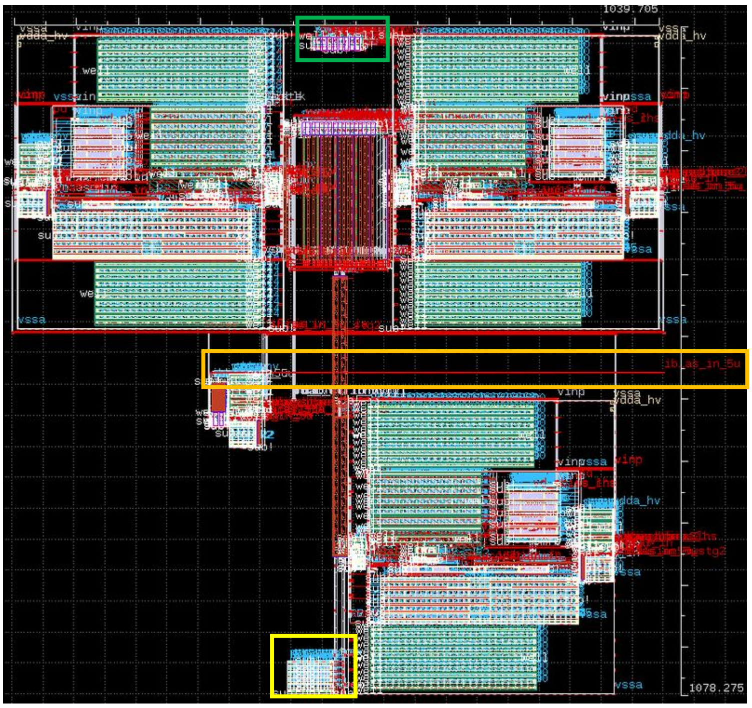

Figure 32 Layout view of sunrise_INA_top

Immediately obvious from Figure 32 is the empty area in this 1mm2 footprint. As will be shown in [15], this area gets occupied by the digital + ring oscillator + temperature sensor + level shifters + internal logic.

In addition to the sub-blocks discussed in section 5, INA top level also includes an input mux (green box in Figure 31) and 5pF output filtering cap (yellow box in Figure 31).

Input mux schematic and layout are shown in Figure 33 where it can be seen to be a 4:1 differential in / differential out mux comprising of 8 instances of the Tgate shown in Figure 26. As will be detailed in [15], the final ASIC is capable of reading 2 external differential inputs (one with CDM ESD protection and one without) in addition to the input bias current required for the on-chip temperature sensor. As will be detailed in [15], the 4th differential input is tied to vssa at the top level as not enough pins were available in the final QFN 36 pin package to accommodate it.

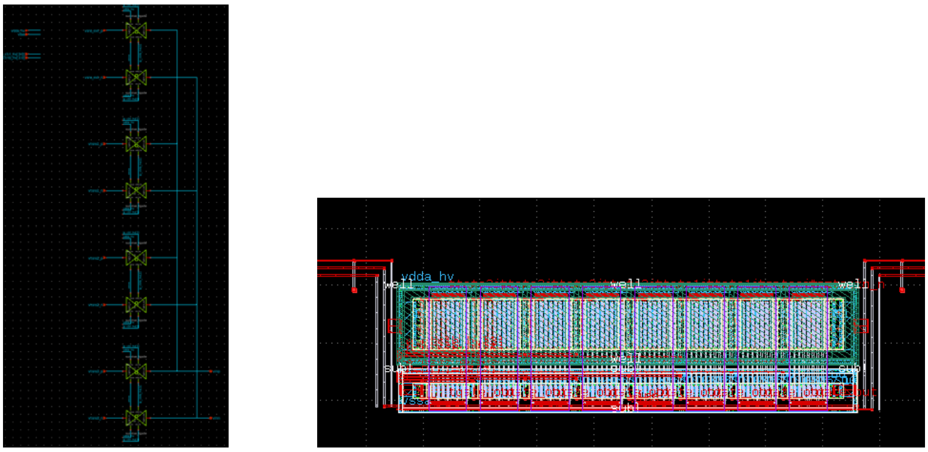

Figure 33 Input mux schematic and layout

Output filtering cap schematic and layout are shown in Figure 34 whose purpose is to prevent noise integrating up on vout, so that the 200uV output integrated noise spec can be met.

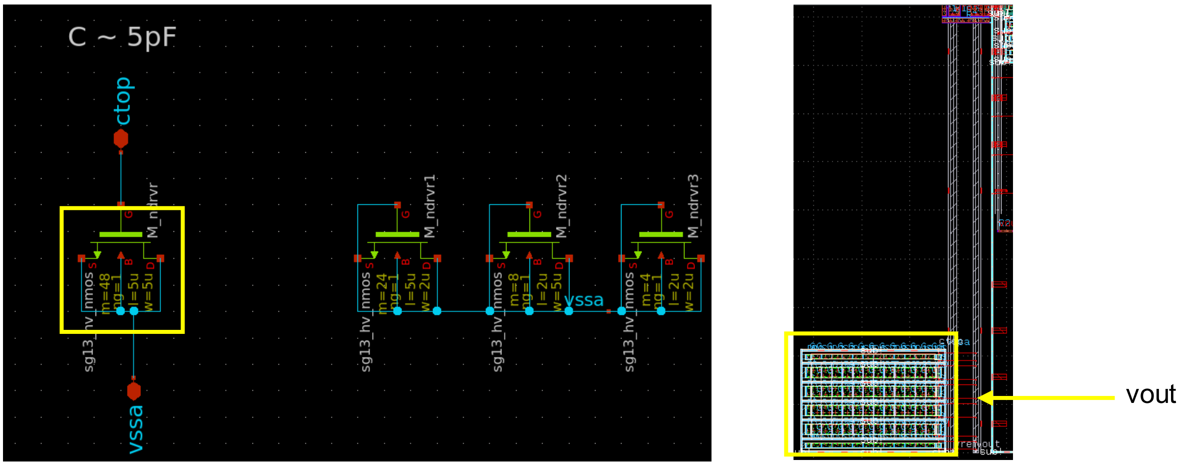

Figure 34 Output filter cap schematic and layout

One final point worth mentioning is the 5uA input bias line shown by the orange box in Figure 32, which is further highlighted in Figure 33. As will be seen in [15], the external 5uA pin exists on the east side of the ASIC which, as shown in Figure 33, travel 730um on a 2um wide line to the input bias block. This line thus represents 365 squares, which at a sheet resistance of $R_s = 0.1\Omega/sq$, gives 36.5Ω routing resistance. Passing a 5uA current across this incurs a 182.5uV IR, i.e. is a negligible amount.

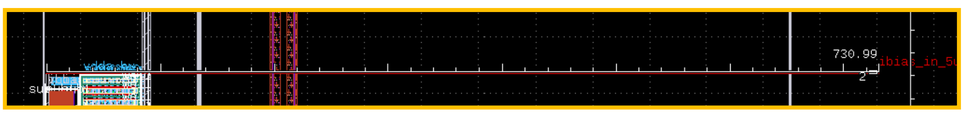

Figure 35 5uA input bias line

6.2 Gain seting = 5 (Av = 5)

6.2.1 Gain error (GE)

GE for an output swing from 50mV → vdda_hv – 50mV is summarized in Table 13 and plotted in Figure 36.

Parameter	SCH	PEX	Delta
GE (%)	-0.12	-0.09	-0.03

Table 13 Gain error summary

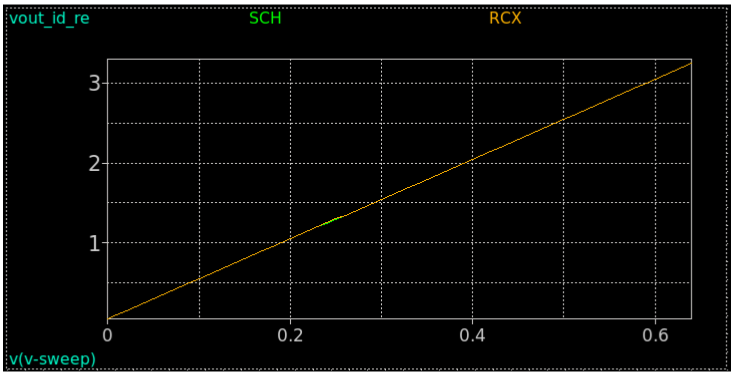

Figure 36 Pre (SCH) and post (RCX) layout GE plots

As can be seen, pre and post layout GE results match very closely and meet the 0.5% top level spec.

6.2.2 Noise density

Input referred noise density at 1kHz is summarized in Table 14 with its corresponding output referred noise density plotted in Figure 37.

Parameter	SCH	PEX	Delta
vn_1kHz (V/rtHz)	50.78	50.78	0

Table 14 Noise density summary

Close observation of Figure 37 will show the PEX waveform to be labelled “CX” instead of the standard “RCX” used for all previous PEX waveforms. This is to indicate that the “CX” waveform is taken from a Conly PEX netlist (whereas “RCX” corresponds to an RC PEX netlist). The reason a Conly netlist was used was because the RC netlist would not converge for ac noise analysis at the top level (an issue not uncommon for such analysis on large top level netlists). The migration to Conly results in no gaps in the verification flow since the noise specs are AC and thus incur negligible impact from the R component of a PEX netlist.

As can be seen, pre and post layout noise density results match and meet the 50nV/rtHz top level spec.

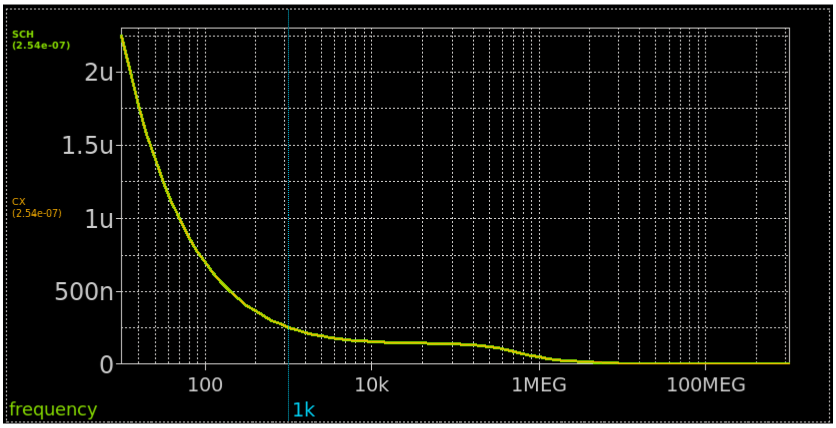

Figure 37 Pre (SCH) and post (CX) layout noise density plots

6.2.3 Integrated noise

Integrated noise is summarized in Table 15 and plotted in Figure 38.

Parameter	SCH	PEX	Delta
vnrms (uV)	113	113	0

Table 15 Integrated noise summary

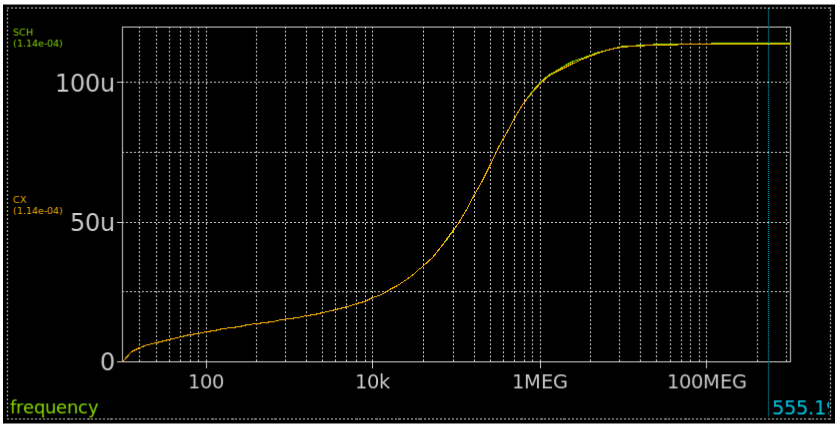

Figure 38 Pre (SCH) and post (CX) layout integrated noise plots

Again, close observation of Figure 38 show a Conly "CX" PEX netlist to be used due to ac noise convergence issues, as explained in section 6.2.2.

As can be seen, pre and post layout noise density results match and meet the 200uVrms top level spec. It should be noted, it is the inclusion of the 5pF shown in Figure 34 that achieves this since it prevents noise continuously integrating up on the output.

6.2.4 CMRR

CMRR is summarized in Table 16 and plotted in Figure 39.

Parameter	SCH	PEX	Delta
CMRR (dB)	-114.4	-114.6	0.2

Table 16 CMRR summary

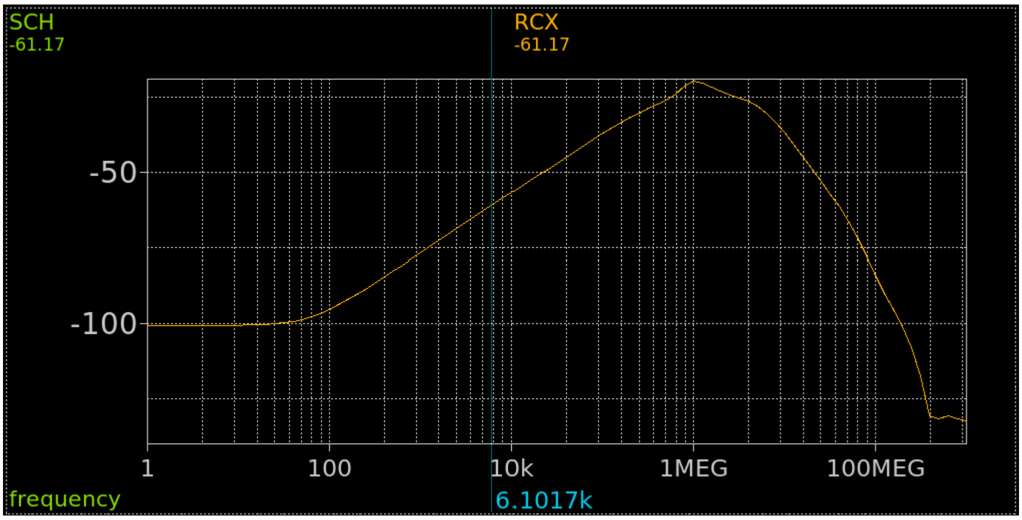

Figure 39 Pre (SCH) and post (RCX) layout CMRR plots

As can be seen, pre and post layout noise density results match exactly. This is a major achievement given the amount of post layout debug required, as mentioned in section 5.3.1 and detailed on slide 14 of [14].

It should be noted that only systematic CMRR was analysed post layout. Top level spec is 80dB, which as described in [13], is dominated by the random component of CMRR (stage 2 resistor mismatch). As such, testing the random component of CMRR requires 30-100 MC runs, which for this large top level netlists represents a major computational overhead and effort. Since MC will not be impacted post layout, this effort was not justified. Instead, as shown, we focused on the systematic component of CMRR since that is what gets impacted (and indeed did get impacted) by post layout parasitics.

In order to meet the 80dB top level spec, it is necessary that the random component dominates, which for post layout analysis, translates to verifying the systematic component is << -80dB (which it is by ~ 35dB i.e. the systematic component is almost 100x less than the random one).

6.2.5 BW

Bandwidth (BW) is summarized in Table 17 and plotted in Figure 40.

Parameter	SCH	PEX	Delta
BW (kHz)	301.4	300.2	1.2

Table 17 BW summary

As can be seen, pre and post layout noise density results match and meet the 100kHz top level spec.

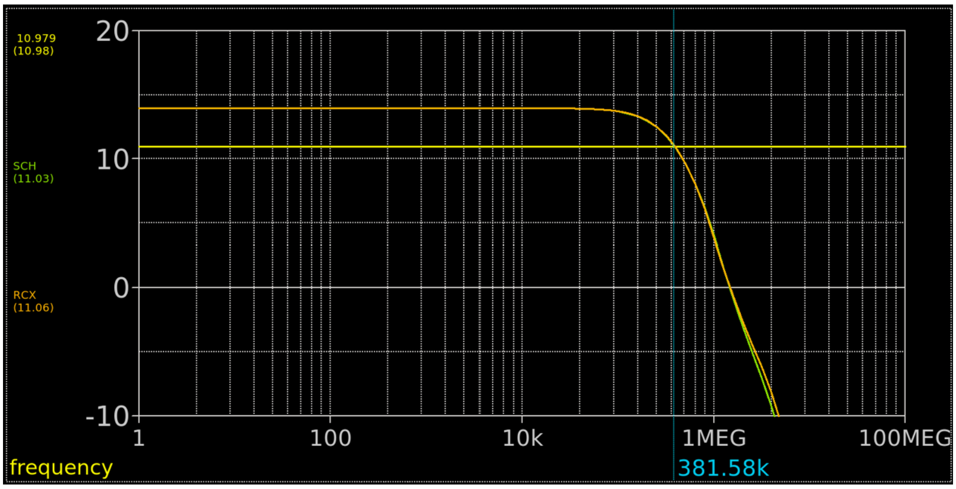

Figure 40 Pre (SCH) and post (RCX) layout BW plots

6.2.6 Stability

As described in [13], due to the multiple loops of an INA, testing its stability is most accurately achieved by examining its step response (10mV step at the INA input). Figure 41 plots the step response pre and post layout.

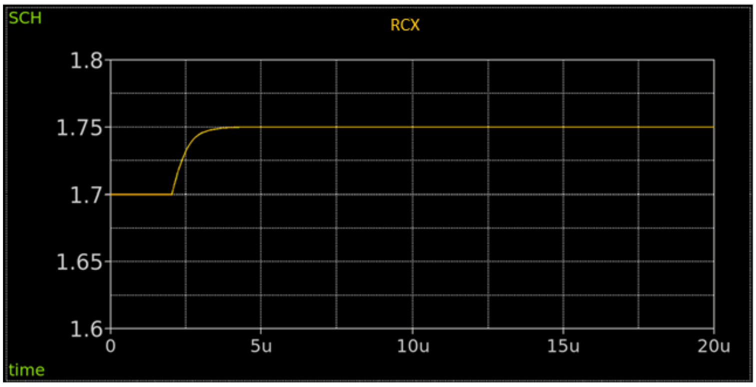

Figure 41 Pre (SCH) and post (RCX) layout step response plots

As can be seen, pre and post layout step response results match where absolutely no ringing occur to verify the INA to be stable post layout.

6.2.7 idda_hv

idda_hv is summarized in Table 18.

Parameter	SCH	PEX	Delta
idda_hv (uA)	786	786	0

Table 18 idda_hv summary

As can be seen, pre and post layout results match and meet the 1mA top level spec.

6.2.8 Post layout summary of the top level

Pre and post layout analysis for the top level is summarized in Table 19.

Parameter	SCH	PEX	Delta
GE (%)	-0.12	-0.09	-0.03
vn_1kHz (V/rtHz)	50.78	50.78	0
vnrms (uV)	113	113	0
CMRR (dB)	-114.4	-114.6	0.2
BW (kHz)	301.4	300.2	1.2
idda_hv (uA)	786	786	0

Table 19 Pre and post layout summary for the INA top level

As can be seen, thanks to the various layout iterations, pre and post layout performances match almost exactly. From this we infer that, since pre layout performance (shown in [13]) met the top level specifications of Table 1 across PVT conditions / INA gain settings, the post layout performance also will (since it displays near identical performance under typical conditions for a gain of 5).

7 References

[1] https://www.analog.com/media/en/technical-documentation/data-sheets/AD8223.pdf

[2] https://ngspice.sourceforge.io/download.html

[3] https://xschem.sourceforge.io/stefan/xschem_man/install_xschem.html

[4] https://openvaf.semimod.de/

[5] https://octave.org/download

[6] https://www.klayout.de/build.html

[7] https://opencircuitdesign.com/magic/download.html

[8] https://opencircuitdesign.com/netgen/

[9] https://github.com/IHP-GmbH/ihp-sg13cmos5l

[10] https://github.com/IHP-GmbH/IHP-Open-PDK

[11]https://www.ihp-microelectronics.com/services/research-and-prototyping-service/mpw-prototyping-service/schedule-price-list

[12] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_1

[13] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_2/INA

[14]https://wiki.f-si.org/index.php?title=Design_of_an_industry_quality_instrumentation_amplifier_using_OS_tools_/_silicon

[15] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_4

8 Revision History

Rev.	Date	Author	Changes, Remarks
A	08.08.2026	Diarmuid Collins	Created

Table 20: Revision history